



Statistics of Financial Markets

Seminar 2: Classical distributions and stylised facts



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Acronyms used in this material

- ▣ ACF: AutoCorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ CDF: Cumulative Distribution Function
- ▣ CLT: Central Limit Theorem
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ JB: Jarque–Bera (test)
- ▣ MLE: Maximum Likelihood Estimation
- ▣ OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))
- ▣ QQ: Quantile–Quantile (plot)
- ▣ SE: Standard Error

Today's Question and Route

- **Question:** how far are real daily returns from the Normal distribution, and how do we measure it?
 - ▶ this seminar comes **before** Lecture 2: the section "What You Need for Today" gives every definition the tasks use
- Route
 - ▶ Part A: Normal, lognormal and Student- t computations and derivations on paper
 - ▶ Part B: moments, tests, QQ plots and stylised facts on real data, each with an interpretation question
 - ▶ Part C: an open question for a project and an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section
- Nothing is handed in: the seminar is for practice; the solutions of [Proposed] tasks are discussed in class

Exercise Map

Task	Question	Type	Model
A1, A2	how likely is a large daily loss under the Normal distribution?	Solved, Proposed	A1
A3, A4	what is the price after one and ten years under lognormal prices?	Solved, Proposed	A3
A5, A6	when is a sum approximately Normal (CLT)?	Solved, Proposed	A5
A7, A8	the Jarque–Bera statistic and the moments of a Student- t	Solved, Proposed	A7
B1, B2	how non-Normal are daily returns, and how precise is the kurtosis?	Solved, Proposed	B1
B3, B4	does a Student- t fit better? how many 4-sigma days?	Solved, Proposed	B3
B5–B7	aggregational Gaussianity, volatility clustering, leverage and asymmetry	Solved, Proposed	B5
C1, C2	are Bitcoin's tails getting thinner? what is wrong in an AI answer?	Proposed	B3, B5

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data Used

Series	Source	Price used	Period
S&P 500, DAX, BET	EODHD	close	2010–2026
Bitcoin	EODHD	close, 7 days a week	2014–2026
OMV Petrom (SNP)	EODHD	adjusted close	2010–2026

- Daily log returns in %: $r_t = 100 (\ln P_t - \ln P_{t-1})$
- Weekends and repeated holiday closes are dropped (except Bitcoin); each series on its own calendar
- In the notebook: `returns('sp500')`, `moments(r)`, `t_fit(r)`; no account or key is needed

What You Need for Today (1/4): the Normal Distribution

- $X \sim N(\mu, \sigma^2)$: density $f(x) = (\sigma\sqrt{2\pi})^{-1} \exp(- (x - \mu)^2 / (2\sigma^2))$, mean μ , variance σ^2
- Standardise: $Z = (X - \mu) / \sigma \sim N(0, 1)$; $P(X \leq x) = \Phi((x - \mu) / \sigma)$, Φ = standard Normal CDF (cumulative distribution function)
 - ▶ useful values: $\Phi(-1.645) = 0.05$, $\Phi(-1.96) = 0.025$, $\Phi(-2.326) = 0.01$; $P(|Z| > 4) = 6.3 \times 10^{-5}$
- Quantile of order α : $x_\alpha = \mu + \sigma z_\alpha$, with $z_\alpha = \Phi^{-1}(\alpha)$
- A **k-sigma day**: $|r_t - \bar{r}| > k s$, with $\bar{r}$ and s the sample mean and standard deviation
- Expected number of **k-sigma days** in n days: $n \cdot P(|Z| > k)$; the count is approximately Poisson with this mean

What You Need for Today (2/4): Lognormal Prices and the CLT

- Y is **lognormal**, $Y \sim LN(m, s^2)$, if $\ln Y \sim N(m, s^2)$
 - ▶ median e^m , mean $e^{m+s^2/2}$, $P(Y \leq y) = \Phi((\ln y - m)/s)$
- If yearly log returns are i.i.d. $N(\mu, \sigma^2)$: $P_T = P_0 e^{r_1 + \dots + r_T}$ with $\ln(P_T/P_0) \sim N(T\mu, T\sigma^2)$
- **CLT** (central limit theorem): for i.i.d. X_i with mean μ and finite variance σ^2 , $X_1 + \dots + X_n \approx N(n\mu, n\sigma^2)$
 - ▶ binomial $B(n, p)$: $P(X \geq k) \approx 1 - \Phi((k - 0.5 - np)/\sqrt{np(1-p)})$ (continuity correction)
- The three conditions: independence, identical distribution, finite variance

What You Need for Today (3/4): Shape, Tests and the Student- t

- Sample skewness $S = m_3/m_2^{3/2}$, sample excess kurtosis $K = m_4/m_2^2 - 3$, with $m_k = \frac{1}{n} \sum_t (r_t - \bar{r})^k$; Normal distribution: $S = K = 0$
 - ▶ under normality, standard errors $\sqrt{6/n}$ and $\sqrt{24/n}$
- **Jarque–Bera**: $JB = \frac{n}{6}(S^2 + K^2/4)$; under normality $\chi^2(2)$; reject at 5% if $JB > 5.99$
- **QQ plot**: sorted data $r_{(i)}$ against model quantiles $F^{-1}((i - 0.5)/n)$; an S-shape means heavier tails than the model
- **Student- $t(\nu)$** : variance $\nu/(\nu - 2)$ for $\nu > 2$, excess kurtosis $6/(\nu - 4)$ for $\nu > 4$; for returns $r = m + sT$
 - ▶ fitted by MLE (maximum likelihood estimation); compared with the Normal distribution by $AIC = 2k - 2\ell$ (lower is better)

What You Need for Today (4/4): Stylised Facts

- **Stylised facts Cont (2001)**: statistical properties shared by most assets and periods
- **ACF**: $\hat{\rho}(h) = \text{Corr}(r_t, r_{t-h})$; 95% band under independence $\pm 1.96/\sqrt{n}$
 - ▶ **Ljung–Box**: $Q(m) = n(n+2) \sum_{h=1}^m \hat{\rho}(h)^2 / (n-h)$, compared with $\chi^2(m)$; $\chi_{0.95}^2(10) = 18.31$
- No linear autocorrelation: $\hat{\rho}(h)$ of r_t close to 0; **volatility clustering**: $\hat{\rho}(h)$ of $|r_t|$ positive and slowly decaying
- **Aggregational Gaussianity**: h -day returns (sums of h daily log returns) are closer to the Normal distribution as h grows
- **Leverage effect**: $L(k) = \text{Corr}(r_t, |r_{t+k}|) < 0$; **gain/loss asymmetry**: large losses bigger and more frequent than large gains
- **Bootstrap**: resample the days with replacement many times, recompute the statistic, take the 2.5% and 97.5% percentiles

A1: A Large Daily Loss under the Normal Distribution [Solved]

Task

- Daily log returns of an index are $N(\mu, \sigma^2)$ with $\mu = 0.05\%$ and $\sigma = 1.1\%$; there are 252 trading days a year.
- 1. Compute the probability of a daily loss larger than 2% and the expected number of such days per year.
- 2. Compute the 1% quantile of the daily return.
- 3. Compute the expected number of 4-sigma days in 10 years.
- Report: a probability, a number of days, a quantile and an expected count.

Solution

- 1. $z = (-2 - 0.05)/1.1 = -1.86$;
 $P = \Phi(-1.86) = 3.12\%$; $252 \times 3.12\% = 7.9$ days a year
- 2. $x_{0.01} = 0.05 + 1.1 \times (-2.326) = -2.51\%$
- 3. $P(|Z| > 4) = 6.3 \times 10^{-5}$; in 2520 days: 0.16 days, so most decades would see none
- Lecture 2 shows that real indices have one such day every few months.

A2: The Same Questions for Bitcoin [Proposed]

Solution discussed in class

Task

- Daily log returns of Bitcoin are $N(0.12\%, 3.6\%^2)$; Bitcoin trades 365 days a year. Model: A1.
- 1. Compute the probability of a daily loss larger than 10% and the expected number of such days per year.
- 2. Compute the 1% quantile of the daily return.
- 3. Compute the expected number of 4-sigma days in 10 years.
- Report: a probability, a number of days, a quantile and an expected count.

A3: Lognormal Prices after One and Ten Years [Solved]

Task

- ▣ A stock costs 100 lei; its yearly log returns are i.i.d. $N(0.07, 0.18^2)$.
- ▣ 1. Compute the median and the mean of the price after one year.
- ▣ 2. Compute the probability that the price after one year is below 100 lei.
- ▣ 3. Compute the 5% quantile of the price after one year.
- ▣ 4. Repeat 1–3 for ten years.
- ▣ Report: eight numbers and one sentence on the horizon.

Solution

- ▣ 1. Median $100e^{0.07} = 107.25$; mean $100e^{0.07+0.0162} = 109.00$
- ▣ 2. $P = \Phi(-0.07/0.18) = \Phi(-0.389) = 34.9\%$
- ▣ 3. $100e^{0.07+0.18 \times (-1.645)} = 79.77$
- ▣ 4. $T = 10$: $10\mu = 0.7$, $\sqrt{10}\sigma = 0.569$; median 201.38, mean 236.79, $P(\text{loss}) = 10.9\%$, 5% quantile 78.96
- ▣ A loss becomes less likely with time, but the bad 5% case barely improves.

A4: Why the Mean Is $e^{m+s^2/2}$ [Proposed]

Solution discussed in class

Task

- Let $X \sim N(m, s^2)$ and $Y = e^X$. Model: A3; exercise of the type in [Borak, Härdle & López-Cabrera \(2013\)](#).
- 1. Write $E[e^X] = \int e^x f(x) dx$ and complete the square in the exponent to show that $E[e^X] = e^{m+s^2/2}$.
- 2. Use the same idea for $E[e^{2X}]$ and derive $\text{Var}(Y) = (e^{s^2} - 1)e^{2m+s^2}$.
- 3. For $m = 0.07$, $s = 0.18$, compute the mean and the median of Y and their gap.
- Report: the two derivations and one sentence linking the gap to the volatility drag of [Chapter 1](#).

A5: Up Days in a Year (Normal Approximation) [Solved]

Task

- ▣ An index rises on a given day with probability $p = 0.53$, independently across days; a year has 252 trading days.
- ▣ 1. Give the distribution of the number X of up days, its mean and its standard deviation.
- ▣ 2. Approximate $P(X \geq 140)$ with the Normal distribution and the continuity correction.
- ▣ 3. Compare with the exact binomial value from the notebook.
- ▣ Report: two moments, two probabilities, one sentence on the quality of the approximation.

Solution

- ▣ 1. $X \sim B(252, 0.53)$; $E[X] = 133.56$,
 $\text{sd} = \sqrt{252 \times 0.53 \times 0.47} = 7.92$
- ▣ 2. $z = (139.5 - 133.56)/7.92 = 0.750$;
 $P \approx 1 - \Phi(0.750) = 22.67\%$
- ▣ 3. Exact: 22.69%
- ▣ $np(1 - p) \approx 63$ is large, so the CLT works very well here.

A6: Is a Monthly Return Normal? [Proposed]

Solution discussed in class

Task

- Daily log returns are i.i.d. with mean 0.04% and standard deviation 1.2%. Model: A5.
- 1. Give the approximate distribution of the 21-day log return and its two parameters.
- 2. Compute the probability of a 21-day loss larger than 10%, and of a 5-day loss larger than 10%.
- 3. Say whether the CLT still applies if the daily returns are Student- t with $\nu = 3$.
- 4. Say whether it applies with $\nu = 1.5$.
- Report: two distributions, two probabilities, two answers with a reason each.

A7: Jarque–Bera on Paper [Solved]

Task

- A series of $n = 4000$ daily returns has sample skewness $S = -0.6$ and excess kurtosis $K = 12$.
- 1. Compute the standard errors of S and K under normality and the two z statistics.
- 2. Compute JB and the share of it that comes from the kurtosis.
- 3. Decide at the 5% level.
- Report: two standard errors, two z values, JB, the decision.

Solution

- 1. $\sqrt{6/4000} = 0.0387$, $\sqrt{24/4000} = 0.0775$; $z_S = -15.5$, $z_K = 154.9$
- 2. $JB = \frac{4000}{6}(0.36 + 36) = 240 + 24\,000 = 24\,240$, almost all from K
- 3. $JB > 5.99$: reject normality
- JB is the sum of the two squared z statistics: $z_S^2 + z_K^2$.

A8: Moments of the Student- t [Proposed]

Solution discussed in class

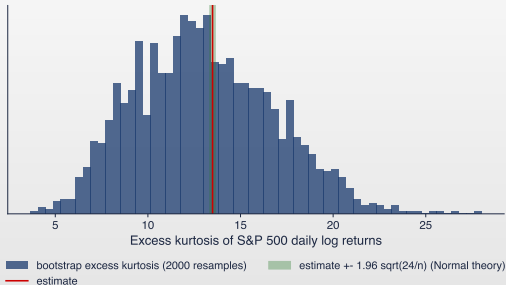
Task

- ▣ $T \sim t(\nu)$; a model for daily returns is $r = m + s T$. Model: A7.
- ▣ 1. Compute the variance and the excess kurtosis of $t(5)$ and $t(10)$.
- ▣ 2. Find ν such that the excess kurtosis is 3.
- ▣ 3. Find the scale s that gives a daily standard deviation of 1.2% for $\nu = 5$ and for $\nu = 6$.
- ▣ 4. Compare $P(|X| > 4)$ for a unit-variance $t(5)$ (value in the notebook) with the Normal value.
- ▣ Report: four moments, one ν , two scales and one ratio.

B1: How Non-Normal Is the S&P 500? [Solved]

- ▣ **Question:** how far are the daily log returns of the S&P 500 from the Normal distribution, and how precisely do we know it?
- ▣ **Inputs:** S&P 500 closes, 2010–2026, daily log returns in %
- ▣ **Tasks**
 1. Compute the mean, standard deviation, skewness, excess kurtosis and the worst day with its date.
 2. Compute the Normal-theory standard errors of S and K , and the JB statistic.
 3. Bootstrap S and K with 2000 resamples of the days and take the 2.5% and 97.5% percentiles.
 4. Interpretation: why is the bootstrap interval of K so much wider than the Normal-theory one?
- ▣ **Report:** a table of moments, JB, two intervals for K , one sentence
- ▣ **Notebook:** section B1

B1: Solution [Solved]



- $n = 4\,201$; mean 0.045%, s.d. 1.08%; worst day -12.8% on 16 March 2020
- $S = -0.61$ (SE 0.038), $K = 13.49$ (SE 0.076); JB = 32 113: normality rejected
- Normal theory: $K \in [13.34, 13.64]$; bootstrap: $[6.7, 20.6]$; for S , bootstrap $[-1.54, 0.29]$
- Interpretation: $\sqrt{24/n}$ assumes Normal data; with heavy tails K depends on a few days, and whether they are resampled moves it a lot; the bootstrap s.d. of K is 50 times larger
- The skewness interval contains 0: the sign of S is not certain

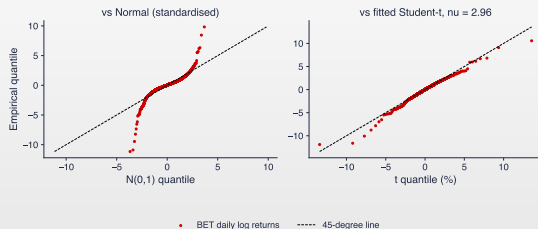
B2: Three More Series [Proposed]

- **Question:** which of the BET, Bitcoin and OMV Petrom is farthest from the Normal distribution, and how much does one day matter?
- **Inputs:** daily log returns, 2010–2026 (Bitcoin from 2014), each on its own calendar; model: B1
- **Tasks**
 1. Compute n , mean, standard deviation, S , K and JB for each series.
 2. Remove the single largest absolute return of each series and recompute K .
 3. Interpretation: what does the change in K tell you about kurtosis as a summary of risk?
- **Report:** one table and two sentences
- **Notebook:** section B2

B3: Normal or Student- t for the BET? [Solved]

- **Question:** does a Student- t describe the daily returns of the BET better than the Normal distribution, and where does each fail?
- **Inputs:** BET closes, 2010–2026, daily log returns in %
- **Tasks**
 1. Draw the QQ plot of the standardised returns against $N(0, 1)$.
 2. Fit a Student- t by maximum likelihood and draw the QQ plot against the fitted t .
 3. Compare the log-likelihoods and the AIC of the two models.
 4. Interpretation: where does the Normal distribution fail, and where does the fitted t fail?
- **Report:** $\hat{\nu}$, $\hat{m}$, $\hat{s}$, two AIC values, two QQ plots, one sentence
- **Notebook:** section B3

B3: Solution [Solved]



- $\hat{\nu} = 2.96$, $\hat{m} = 0.078\%$, $\hat{s} = 0.628\%$; implied s.d.
 $\hat{s}\sqrt{\hat{\nu}/(\hat{\nu} - 2)} = 1.10\%$
- $\ell_t = -5496$ vs $\ell_N = -6224$; $AIC_N - AIC_t = 1454$: the t wins clearly
- Lowest return -11.9% ; matching quantile: Normal distribution -3.9% , fitted t -13.4%
- Interpretation: the Normal distribution misses both tails (S-shape); the t fits the centre and most of the tails, but its extreme quantiles go slightly beyond the data

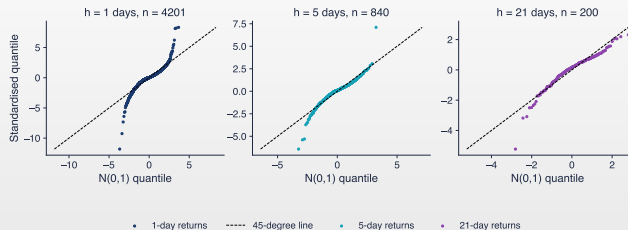
B4: How Many 4-Sigma Days? [Proposed]

- ▣ **Question:** how many 4-sigma days do the S&P 500, DAX, BET and Bitcoin have, and which model predicts the count?
- ▣ **Inputs:** daily log returns, 2010–2026 (Bitcoin from 2014); model: B3
- ▣ **Tasks**
 1. Count the days with $|r_t - \bar{r}| > 4s$ in each series.
 2. Compute the expected count under the Normal distribution and the Poisson probability of seeing at least the observed count.
 3. Compute the expected count under the fitted Student- t (same threshold $\bar{r} \pm 4s$).
 4. Interpretation: is the Student- t too heavy, too light or about right for each series?
- ▣ **Report:** one table and one sentence per series
- ▣ **Notebook:** section B4

B5: Does the S&P 500 Become Normal at Longer Horizons? [Solved]

- **Question:** are 5-day and 21-day returns of the S&P 500 closer to the Normal distribution than daily returns?
- **Inputs:** S&P 500 daily log returns, 2010–2026; h -day returns are sums over non-overlapping blocks of h days
- **Tasks**
 1. Build the 5-day and 21-day log returns.
 2. Compute n , S , K , the standard error $\sqrt{24/n}$ and JB with its p -value at $h = 1, 5, 21$.
 3. Draw the three QQ plots against $N(0, 1)$.
 4. Interpretation: is the monthly distribution Normal, and why does the test have less power at $h = 21$?
- **Report:** a table with three rows, three QQ plots, two sentences
- **Notebook:** section B5

B5: Solution [Solved]



h	n	S	K	$\sqrt{24/n}$	JB (p)
1	4201	-0.61	13.49	0.08	32 113 (≈ 0)
5	840	-0.62	7.06	0.17	1 797 (≈ 0)
21	200	-1.18	3.68	0.35	159 (2.9×10^{-35})

- Interpretation: K falls from 13.49 to 3.68, but monthly returns are still not Normal (skewness -1.18 , March 2020); with only 200 months, the standard error of K is 0.35, so smaller departures would go undetected

B6: Are BET Returns Independent? [Proposed]

- **Question:** are the daily returns of the BET uncorrelated, and are they independent?
- **Inputs:** BET daily log returns, 2010–2026; model: B5
- **Tasks**
 1. Compute the ACF of r_t and of $|r_t|$ at lags 1–10 and the band $\pm 1.96/\sqrt{n}$.
 2. Compute the Ljung–Box statistic $Q(10)$ for r_t and for $|r_t|$ and compare with $\chi^2_{0.95}(10)$.
 3. Interpretation: can returns be uncorrelated but not independent?
- **Report:** two ACF columns, two Q values, one sentence
- **Notebook:** section B6

B7: Is Bitcoin Like a Stock Index? [Proposed]

- **Question:** do the S&P 500 and Bitcoin both show the leverage effect and the gain/loss asymmetry?
- **Inputs:** daily log returns, S&P 500 2010–2026, Bitcoin 2014–2026; model: B5
- **Tasks**
 1. Compute $L(k) = \text{Corr}(r_t, |r_{t+k}|)$ for $k = 1, \dots, 5$ and their mean, with the band $\pm 1.96/\sqrt{n}$.
 2. Count the days below $\bar{r} - 4s$ and above $\bar{r} + 4s$; test whether falls and rises are equally likely (binomial test, $p = 0.5$).
 3. Compare $|q_{0.01}|$ and $q_{0.99}$ and the skewness.
 4. Interpretation: which stylised facts of stock indices does Bitcoin share?
- **Report:** one table and two sentences
- **Notebook:** section B7

C1: Are Bitcoin's Tails Getting Thinner? [Proposed]

- ▣ **Question:** has the tail of Bitcoin's daily returns become thinner since 2015, as its market matured?
- ▣ **Inputs:** Bitcoin and S&P 500 daily log returns, 2015–2026; models: B3, B5
- ▣ **Tasks**
 1. Fit a Student- t and compute the excess kurtosis for each calendar year and each series.
 2. Compute the Spearman rank correlation between the year and $\hat{\nu}$.
 3. Fit one t on 2015–2020 and one on 2021–2026 and compare $\hat{\nu}$.
 4. Propose one way to make the answer more reliable (intervals, other windows, other crypto assets).
 5. Interpretation: what can 12 yearly estimates tell us about a trend?
- ▣ **Report:** a table, the rank correlation with its p -value, a short plan for a project
- ▣ **Notebook:** section C1

C2: Audit an AI Answer [Proposed]

- A student asked an AI assistant: "Describe the distribution of daily S&P 500 returns in 2010–2026." The answer:
- (a) The excess kurtosis is 13.5; since the Normal distribution has kurtosis 0, the kurtosis of the S&P 500 is 13.5.
- (b) $JB = 32113$ with $p = 0.000$, which proves that the returns follow a Student-t distribution.
- (c) A Normal 4-sigma day happens once every 15787 days; the 28 such days in 4201 days are just bad luck.
- (d) The lag-1 autocorrelation is only -0.11 , so returns are independent and volatility cannot be predicted.
- (e) With yearly log returns $N(7\%, 18\%^2)$, the expected value of 100 invested after one year is $100 \exp(0.07) = 107.25$.
- **Tasks**
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and the correct number from the notebook (section C2).
 - ▶ 2. Report: a list of five verdicts with one line of justification each.

What You Should Take from Today

- ▣ The Normal distribution makes 4-sigma days almost impossible; real markets have them every few months
- ▣ Lognormal prices: the median is below the mean by the volatility drag
- ▣ Skewness and kurtosis are fragile: report bootstrap intervals and the extremes
- ▣ The Student- t fits much better than the Normal distribution, but not perfectly
- ▣ Returns are nearly uncorrelated but not independent: their size is predictable
- ▣ An AI answer is a draft: check every definition, model and number

After the Seminar

- Lecture 2 develops each topic of today: the Normal distribution and lognormal models, the CLT, moments and tests, the Student- t and six stylised facts
- Try the [Proposed] tasks in the notebook; the solutions are discussed in class
- C1 can grow into a team project: rolling windows, bootstrap intervals, more crypto assets
- Reading: [Franke, Härdle & Hafner \(2019\)](#), Sec. 3.3 and Sec. 11.2; exercises with solutions: [Borak, Härdle & López-Cabrera \(2013\)](#)

References

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